

QUICKFLOW: ROBUST DESIGN, MATERIALITY, AND CMF PROPOSAL

Project Context: Portable Adaptable Hydration System Ecosystem

Brand Collaboration: UNBROKEN

Authors: Andrea Díaz de León Aguilar (A01613127), Juan Pablo Cedillo Pérez (A01713855)

Date: May 21, 2026

TABLE OF CONTENTS

- 1. Material Strategy & Lifecycle Paradigms
- 2. Comprehensive Component Breakdown & Bill of Materials (BOM)
- 3. Engineered Material Properties & Performance Benchmarks
- 4. Manufacturing & Production Process Mapping
- 5. Color, Materials, and Finish (CMF) Specification Palette
- 6. Material Failure Modes, Risks, and Mitigation Strategies
- 7. Alternative Material Explorations for Future Product Generations
- 8. Technical Report Language & Engineering Specifications
- 9. Commercial & Visual Marketing Copy Translation
- 10. Product Development Summary & Protocol Stepping

1. MATERIAL STRATEGY & LIFECYCLE PARADIGMS

The design philosophy of Quickflow moves away from seeing personal water containers as single, unchangeable, throwaway shapes. Traditional insulated flasks rely on heavy, rigid double-wall steel bodies that can permanently dent, lose their vacuum seal, and end up in landfills. On the other hand, traditional soft flasks run the risk of puncture, collect odors, and lack thermal protection. Developed alongside **UNBROKEN**, Quickflow creates a balanced material framework that brings together two key goals: high-performance sports durability and clean, everyday wellness appeal.

[HIGH-TACTILE PROTECTIVE SLEEVE]	<- Conatex Technical Textile Wrap
[REGULATED THERMAL CONTROL]	<- Kryotec Removable Gel Chamber

[COLLAPSIBLE FLUID EMBODIMENT]		<- RF-Welded TPU Core
Vessel		
[STRUCTURAL UTILITY CORNERSTONE]		<- Injection-Molded
Rigids & Caps		

The Layered Architecture Strategy

To achieve a product that naturally follows user movement without adding unnecessary bulk, the material choice follows a three-layered architecture:

- **The Core Fluid Vessel (Flexible Layer):** The primary water container features a distinct triangular-pyramidal shape. It must hold its shape well when full, but fold down easily when empty to minimize storage volume. It uses a high-flex-life thermoplastic elastomer that stays clear over time, allowing the integrated liquid-level markers to remain easily readable.
- **The Thermal Control Core (Removable Layer):** Instead of using heavy, permanent insulation walls, thermal control is moved to a removable cooling gel pocket. This modular design helps keep the overall weight low when insulation isn't required.
- **The External Protective Textile Frame (Tactile Layer):** This outer layer shields the flexible core from abrasions, manages sweat condensation, houses the quick-release magnetic components, and provides a comfortable grip during high-intensity sports.

Circular Economy Framework

Following the sustainability standards set in the initial design phases, this material plan relies heavily on modular construction. Every key component—from the flexible main body and cooling gel insert to the external fabric sleeve and drinking nozzle—can be taken apart without destructive tools. If a runner accidentally punctures the soft body after extended use, they only need to swap out that specific part instead of throwing away the entire product. This individual parts replacement strategy cuts down on electronic and polymer waste, raises the product's overall value, and lets users mix and match visual accessories over time.

2. COMPREHENSIVE COMPONENT BREAKDOWN & BILL OF MATERIALS (BOM)

The table below breaks down the technical details for each part of the Quickflow ecosystem. Final mechanical dimensions, fluid volumes, and exact weight targets are currently **to be defined** as part of upcoming engineering prototyping phases.

Part ID	Component Name	Primary Candidate Material	Material Sub-Type / Grade	Material Function & Justification
QF-101	Flexible Fluid Chamber	Thermoplastic Polyurethane (TPU)	Polyether-based food-grade film, aromatic or aliphatic <i>[Needs testing]</i>	Forms the collapsible, triangular-pyramidal container. Chosen for high tear resistance, clean taste profile, and excellent RF-welding traits.
QF-102	Rigid Structural Base Ring	Polypropylene (PP)	High-impact injection-molding grade, BPA/BPS-free	Provides a solid bottom frame and reliable mounting threads for the main container assembly.
QF-103	Structural Closure Cap	Polypropylene (PP)	Medical/food-grade, high torque resistance	Seals the top fluid paths and protects internal components from outside debris or drop

Part ID	Component Name	Primary Candidate Material	Material Sub-Type / Grade	Material Function & Justification
				damage.
QF-104	Interchangeable Active Nozzle	Liquid Silicone Rubber (LSR)	Platinum-cured food-grade silicone, high-tear grade	Provides a comfortable self-sealing capability for reliable, hands-free use on runs.
QF-105	Removable Cooling Gel Chamber	Ethylene-Vinyl Acetate (EVA) housing + Phase Change Material	Kryotec proprietary cooling formulation in sealed polymer skin	Placed inside the protective sleeve to absorb heat and keep water cold without adding rigid bulk.
QF-106	Protective External Sleeve	Technical Recycled Polyester (rPET) / Ripstop Nylon	Conatex engineered DWR (Durable Water Repellent) texturized weave	Shields the internal TPU vessel from dirt and damage, absorbs surface condensation, and offers an ergonomic grip.
QF-107	Magnetic Core Elements	Neodymium Iron Boron (NdFeB)	N52 High-flux density rare-earth permanent magnets	Delivers strong, reliable alignment forces for securing the bottle while moving.

3. ENGINEERED MATERIAL PROPERTIES & PERFORMANCE BENCHMARKS

Every selected material must meet specific baseline performance metrics to ensure safety, durability, and a consistent user experience during outdoor and athletic activities.

3.1. Mechanical & Structural Metrics

Polyether-Based TPU Film (QF-101)

- **Hardness Durometer Range:** *[To be defined; target range 80A - 90A needs evaluation]*
- **Ultimate Tensile Strength:** ≥ 35 MPa to withstand high internal pressure from drops or squeeze forces.
- **Elongation at Break:** $\geq 450\%$ to ensure the unique triangular-pyramidal geometry can stretch and flex without tearing.
- **Flex Fatigue Resistance:** Must survive at least 10,000 complete fold-and-unfold cycles under normal conditions without stress-whitening or cracking.

Injection-Molded Polypropylene/Tritan Rigids (QF-102, QF-103)

- **Izod Notched Impact Strength:** ≥ 45 J/m to prevent fracturing if dropped on hard surfaces like concrete or gym floors.
- **Flexural Modulus:** $\geq 1,500$ MPa to maintain reliable screw thread sealing alignment under high tension.

3.2. Thermal & Environmental Profiles

Kryotec Cooling Insert Gel (QF-106)

- **Phase Transition Target Temperature:** Optimized to keep internal liquids near 4°C to 8°C.
- **Latent Heat of Fusion:** *[Needs testing]* to determine how long water stays chilled across outdoor ambient temperatures up to 40°C.
- **Thermal Conductivity Barrier:** The surrounding EVA skin must slow heat transfer down sufficiently to prevent cold-burns during handling.

External Technical Sleeve Textile (QF-107)

- **Hydrophobic Rating:** Grade 4 or higher on AATCC 22 Water Repellency tests, ensuring rainwater rolls off instead of soaking the fabric.
- **Moisture Vapor Transmission Rate (MVTR):** *[To be defined]* to ensure sweat and

ambient condensation evaporate quickly rather than creating damp spots.

3.3. Chemical Safety & Biocompatibility Status

- **Leachability Requirements:** Every water-contact part (QF-101, QF-102, QF-103, QF-104) must remain 100% free of BPA, BPS, phthalates, and heavy metals.
- **Organoleptic Neutrality:** Materials must pass rigorous taste and odor extraction tests, ensuring no plastic or chemical taste transfers into the drinking water after sitting for 24 hours at 40°C.
- **Antimicrobial Performance:** Internal fluid contact parts and external textiles must use permanent antimicrobial defenses to block mold, mildew, and odor causing bacteria from sweat or humid air.

4. MANUFACTURING & PRODUCTION PROCESS MAPPING

Quickflow's production model splits manufacturing across specialized regional production facilities (**IPM, Hi-Tech Products, Kryotec, and Conatex**) to handle advanced welding, molding, and technical fabric assembly.

```
[ TPU Roll Stock Stock ] ---> [ Precision Die-Cutting ] ---> [ RF  
Welding Machine (IPM/Hi-Tech) ] ---+
```

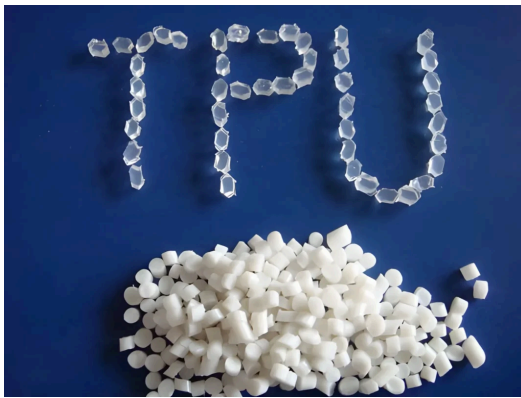
```
|  
[ Polymer Granules ] ---> [ Injection Molding ] ---> [  
Structural Base Components ] ---> [ Final Assembly Room ]
```

```
|  
[ Conatex Fabrics ] ---> [ Automated Stitching ] ---> [  
Magnetic Overmolding/Sleeve ] ---+
```

4.1. Radio Frequency (RF) Welding (Vessel Body Manufacturing)

The manufacturing pipeline for the main fluid vessel uses automated Radio Frequency (RF) welding paths handled by **IPM** and **Hi-Tech Products**.

1. **Material Feeding & Orientation:** Flat sheets of food-grade polyether TPU film are unrolled and fed into position using precise physical registration markers.
2. **Die Cutting:** High-speed machinery punches out the specific flat shapes required to fold into the final three-dimensional triangular-pyramidal form.
3. **High-Frequency Energy Activation:** The cut film pieces are clamped inside custom-shaped brass or aluminum tooling dies. High-frequency electromagnetic energy (27.12 MHz) passes through the joint areas. This excites the polymer molecules, causing them to heat rapidly and fuse into a seamless, pressure-sealed structural bond.
4. **Rigid Frame Bonding:** In the same production line, the flexible TPU body is welded onto the rigid collar ring (QF-102). This process must form a perfect seal without relying on chemical adhesives.
5. **Quality Assessment:** *[Pressure testing thresholds to be defined]*; finished containers are pressure-tested inline to check for microscopic leaks before moving to final assembly.



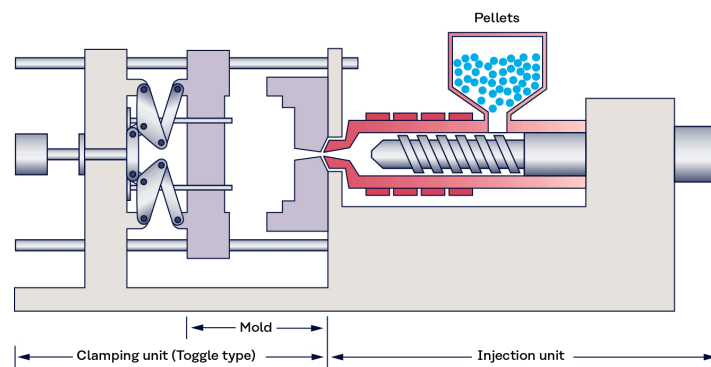
TPU Material, flexibility and our design



4.2. Injection Molding (Rigid Infrastructure Components)

The screw caps, base rings, and magnetic outer covers are produced using standard high-pressure injection molding:

- **Drying Protocol:** Resin pellets are dried to a strict internal moisture content under 0.02% to prevent cosmetic defects or weak spots in production.
- **Process Settings:** Mold temperatures and injection pressures are closely controlled to avoid visible knit lines around critical fluid seals and structural threads.



4.3. Technical Textile Production & Magnetic Integration

How is the external sleeve handled?

1. **Automated Cutting:** Multi-layer ripstop nylon and breath-mesh panels are precisely cut using automated laser systems to prevent edge fraying.
2. **Pocket Assembly:** The encased magnetic parts are sewn directly into structural fabric pockets within the sleeve, reinforced with heavy-duty bartack stitching to handle constant pulling forces.

3. **Stitching Construction:** The panels are assembled using durable flatlock seams to keep the interior surface smooth against the inner bottle core.



5. COLOR, MATERIALS, AND FINISH (CMF) SPECIFICATION PALETTE

The visual style rules match the **IDENTITY BOOK** identity system. This color scheme balances energetic performance accents with a clean, high-end lifestyle look.

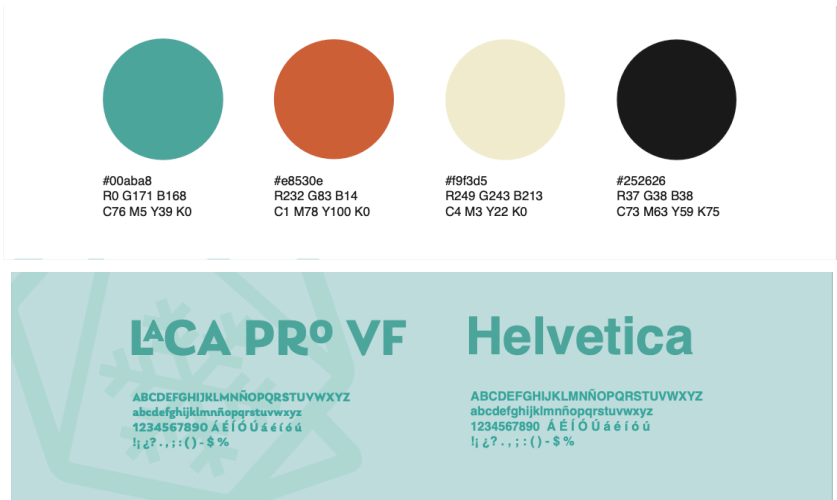
5.1. Definitive Color Reference Table

Color Name	Hex Code	RGB Values	CMYK Percentages	Primary Component Mapping
Brand Core Charcoal	#252626	R37, G38, B38	C73, M63, Y59, K75	Structural caps, rigid base rings, magnetic mounts, and main textile support straps.
Anchor Off-White / Sand	#f9f3d5	R249, G243, B213	C4, M3, Y22, K0	Primary color option for the external technical sleeve and lifestyle-focused product variants.
Energy Coral Pop	#e8530e	R232, G83, B14	C1, M78, Y100, K0	High-visibility interactive points: nozzle tips, cap releases, and alignment markings.

Color Name	Hex Code	RGB Values	CMYK Percentages	Primary Component Mapping
Outdoor Teal Accent	`#00aba8`	R0, G171, B168	C76, M5, Y39, K0	Special edition sleeve patterns, brand logos, and custom accent highlights.

5.2. Texture & Tactility Specifications

- **Main Flexible Fluid Chamber (QF-101):** Fine-grained, chemical-etched semi-frosted translucent finish. Provides a soft, hides surface micro-abrasions, and diffuses light cleanly so users can easily see water levels.
- **Rigid Polymer Assemblies (QF-102, QF-103):** Micro-textured matte finish matching SPI-C1 or Mold-Tech MT-11005 standards. Eliminates distracting plastic shine, matches the premium look of the UNBROKEN brand, and provides reliable grip friction for wet hands.
- **Interactive Trim Elements (QF-104, Accents):** High-gloss polished finish (SPI-A2). Creates a smooth, comfortable mouthfeel on the drinking nozzle and stands out visually to guide the user's eye to interactive points.
- **Technical External Textile Sleeve (QF-107):** Interwoven double-grid ripstop pattern combined with a low-gloss tactile surface finish. Emphasizes high-performance athletic utility, resists fabric pilling against sports apparel, and feels premium to the touch.



6. MATERIAL FAILURE MODES, RISKS, AND MITIGATION STRATEGIES

Developing a modular system that blends flexible films with rigid parts introduces unique technical challenges. The matrix below identifies key material risks and outlines early mitigation plans.

[Critical Stress Joint] ---> Delamination Danger ---> Mitigation: Double-weld paths & radiused anchor lips

[UV Exposure Over Time] ---> Discoloration/Yellowing ---> Mitigation: Introduce non-yellowing aliphatic TPU stabilizers

[Cyclic Folding Motion] ---> Stress-Whitening Fail ---> Mitigation: Polyether TPU optimization & thin wall tests

Component Affected	Hazard / Risk Event	Root Cause Analysis	Design Mitigation Plan
Welded Base Seam (QF-101 to QF-102)	Joint separation or leakage under impact.	Sharp stress spikes where flexible TPU joins the rigid base ring during hard drops.	Add a wide, curved radius to the rigid attachment point to distribute impact forces. Use double-welding tool paths <i>[Weld geometry needs verification]</i> .
Flexible Fluid Chamber (QF-101)	Visual yellowing and material brittleness over time.	Polymer chains breaking down due to sunlight UV exposure and skin oils from hands.	Transition from standard aromatic TPU formulations to specialized aliphatic TPU compounds. These contain built-in UV blockers to keep the container clear <i>[Needs environmental</i>

Component Affected	Hazard / Risk Event	Root Cause Analysis	Design Mitigation Plan
			<i>chamber testing</i>].
Internal Magnetic Core (QF-109)	Rusting and structural swelling of rare-earth magnets.	Tiny pinholes or cracks in the plastic outer casing that let sweat or wash water seep inside.	Use a multi-layer nickel-copper-epoxy protective plating on the magnets before overmolding them in PC/ABS plastic.
Kryotec Inner Skin (QF-106)	Punctures or bursting when frozen.	Liquid expanding inside the gel pack when frozen, putting high stress on flat outer seams.	Switch from simple flat lap seams to a deep-drawn blister cup design with generous expansion zones. <i>[Maximum filling limits need testing]</i> .
Technical Fabric Sleeve (QF-107)	Mildew smell and fabric staining from sweat.	Moisture trapped in the fabric weave from outdoor runs or humid gym bags.	Infuse the fabric fibers with a permanent silver-ion antimicrobial coating during raw yarn production.

7. ALTERNATIVE MATERIAL EXPLORATIONS FOR FUTURE PRODUCT GENERATIONS

To help the Quickflow ecosystem grow, alternative material options are being researched to address sustainability goals and extreme environmental conditions.

7.1. Bio-Based and Circular Polymers (Sustainability Upgrades)

- **Yulex / Bio-Based Neoprene Substitutes:** Future versions could replace traditional polyester textile sleeves with Yulex bio-rubbers sourced from FSC-certified hevea trees. This cuts the carbon footprint of raw sleeve production up to 80% while keeping excellent impact protection and thermal insulation.
- **PHA (Polyhydroxyalkanoates) Rigid:** Injection-molded caps and base rings could use marine-degradable PHA formulations. This ensures that even if a small component is lost outdoors, it will break down safely without creating microplastics.
- **Bio-TPU Materials:** Explore using corn or sugarcane-derived flexible polyurethane resins for the main body. This reduces reliance on oil-based plastics while matching the material strength of standard technical gear.

7.2. Advanced Elastomer Upgrades (High-Durability Variants)

Platinum-Cured Liquid Silicone Rubber (Full Body Substitution): If TPU film shows unexpected fatigue limits during long-term testing, the entire flexible body could be made using compression-molded liquid silicone rubber.

- *Advantages:* Virtually indestructible flex life, completely flavor-neutral, dishwasher safe, and handles freezing temperatures without becoming brittle.
- *Disadvantages:* Silicone cannot be RF-welded, meaning the assembly would require mechanical clamping rings. It also naturally attracts dust and lint, which *[requires specialized anti-static surface treatments]*.

8. TECHNICAL REPORT LANGUAGE & ENGINEERING SPECIFICATIONS

This section provides formal engineering language templates to use in upcoming design verification documents and manufacturing blueprints.

8.1. Material System and Interface Mechanics

"The modular fluid containment architecture utilizes high-frequency radio frequency (RF) welding paths to create a permanent, hermetic bond between the primary polyether-based Thermoplastic Polyurethane (TPU) film body and the structural injection-molded, BPA-free Polypropylene (PP) collar ring assembly. This joint must sustain a minimum internal pressure metric of *[to be defined via hydro-static testing]* without showing signs of structural deformation,

micro-tearing, or seal failure. The fluid path is sealed by a high-compression, food-grade EPDM elastomer ring with a hard durometer target of *[to be defined]* Shore A. This gasket must maintain a reliable liquid seal under a torque range of *[to be defined]* N·m applied to the main cap threads."

8.2. Thermal Regulation and Magnetic Field Enclosure Specifications

"To control thermal properties, the system uses a removable Phase Change Material (PCM) gel pack designed by **Kryotec**. This gel module is placed inside an insulated technical fabric layer manufactured by **Conatex**. The fabric layer uses high-density recycled polyester fibers treated with a long-lasting water repellent layer to prevent condensation buildup. The attachment mechanism relies on a pair of rare-earth Neodymium (NdFeB) N52 magnet arrays. These magnets must deliver a constant holding force of *[to be defined via push-pull gauge]* through the fabric layers. To protect the magnetic components from corrosion caused by salt sweat and moisture, they are completely enclosed inside an overmolded case made of impact-resistant Polycarbonate and ABS plastic."

9. COMMERCIAL & VISUAL MARKETING COPY TRANSLATION

This section translates the technical material data into engaging narrative copy for commercial catalogs, lifestyle lookbooks, website content, and retail packaging.

Technical Data Parameter	Consumer-Facing Benefit Experience
RF-Welded Polyether TPU	"Featherlight, space-saving design folds away completely to eliminate bulky bounce on your run."
Kryotec Thermal Insert	"Keeps your water crisp and ice-cold, without the heavy bulk of old-school metal flasks."
LACA PRO Visual Window	"A clean, simple tracking window keeps you in tune with your daily hydration targets."

9.1. Brand Narrative Pitch (Website & Retail Packaging Copy)

"Meet Quickflow—the advanced hydration system designed to follow your natural pace,

developed in partnership with **UNBROKEN**. Traditional metal flasks are heavy, bulky, and dent easily when dropped. Quickflow changes the game with an adaptable, flexible core that moves with you and folds down small once empty to save space in your pack. Crafted from ultra-durable, food-grade materials that never alter your water's natural taste, Quickflow keeps your water clean, fresh, and cold. The soft, frosted body feels natural in your hand, while bright energy-colored details make it easy to operate while sprinting. Protected by a premium, weather-resistant fabric sleeve, Quickflow feels like a natural extension of your workout gear."

9.2. Retail Catalog Bullet Points (Features & Benefits Breakdown)

- **Dynamic Space-Saving Body:** The durable TPU body keeps a solid triangular shape when full, but folds down flat when empty to eliminate annoying bulk and weight.
- **Advanced Kryotec Cooling:** A removable cooling gel core keeps water ice-cold on hot outdoor runs without adding the weight of thick insulated walls.
- **Quick-Release Magnetic Docking:** Strong built-in magnets let you clip and unclip the bottle from backpacks or running belts instantly without looking down.
- **Clear Hydration Tracking:** A semi-frosted tracking window lets you easily monitor your water level at a glance, helping you reach your daily health goals.
- **Durable Technical Fabric Sleeve:** Designed with high-performance fabrics from **Conatex**, the outer sleeve absorbs sweat condensation, resists scuffs, and gives you a secure, non-slip grip.
- **Clean Taste Guarantee:** Built using 100% BPA-free, medical-grade materials that keep your water completely fresh, clean, and taste-free all day long.

9.3. Brand Copy Matrix: Technical Data to Consumer-Facing Value

Raw Material Terminology	Consumer-Facing Copy	Visual Image Style Direction
Aliphatic Food-Grade Polyether TPU Film with Frosted Surface Treatment.	"A velvety, cloud-soft flexible core that stands up to rugged use, keeps water tasting pure, and won't stain or yellow over time."	A clean, bright close-up macro shot showing water droplets splashing across the textured, frosted bottle body.
Plated N52 Neodymium Magnets Overmolded in	"Click and go. A reliable, snap-to-fit magnetic	An action shot of a runner snapping the bottle onto their

Raw Material Terminology	Consumer-Facing Copy	Visual Image Style Direction
Rigid PC/ABS Plastic Outer Casing.	connection that secures your water to your gear instantly while you run."	pack without looking down, focusing on the secure connection point.
Textured Ripstop rPET Technical Fabrics with Water Repellent Finish from Conatex.	"A tough, sports-ready outer fabric that shrugs off rain, absorbs sweat condensation, and provides a comfortable, confident grip."	A crisp, high-contrast studio photo highlighting the woven fabric patterns against a clean, charcoal background.

10. PRODUCT DEVELOPMENT SUMMARY & PROTOCOL STEPPING

To finalize production drawings and prepare for initial field trials, the missing data points highlighted across this document must be resolved through systematic validation steps:

[Prototype Testing] ---> Evaluate Wall Thickness & Elastomer Shore Hardness

[Thermal Lab Tests] ---> Confirm Kryotec Phase Change Metrics & Cold Retention

[Drop-Test Protocols] ---> Field Validate Welded TPU Seam Margins on Concrete

1. **Elastomer Hardness Validation:** Run user test tests with different TPU shore hardness ratings (80A to 90A) to find the best balance between a comfortable grip and easy folding.
2. **Thermal Performance Profiling:** Test the Kryotec gel formulas inside environmental chambers to measure exactly how many hours water stays cold in high heat.
3. **Drop Testing:** Drop full prototypes from various heights onto concrete surfaces to verify the strength of the welded base joints.
4. **Magnetic Pull Tuning:** Measure and adjust the overmolded magnet housing thickness to find the perfect pull force—secure enough to stay attached while running, but easy to release with one hand.